

Exercise JA

Q1. The coefficients of three consecutive terms of $(1 + x)^{n+5}$ are in the ratio 5 : 10 : 14.
Then $n =$

Ans. 6

Sol. Let the three consecutive terms be

$${}^{n+5}C_{r-1}, {}^{n+5}C_r, {}^{n+5}C_{r+1}$$

$$\therefore \frac{{}^{n+5}C_{r-1}}{{}^{n+5}C_r} = \frac{1}{2} \quad \Rightarrow \quad \frac{r}{n-r+6} = \frac{1}{2}$$

$$\Rightarrow n = 3r - 6 \quad \dots\dots(1)$$

$$\text{Also, } \frac{{}^{n+5}C_r}{{}^{n+5}C_{r+1}} = \frac{5}{7} \quad \Rightarrow \quad \frac{r+1}{n-r+5} = \frac{5}{7}$$

$$\Rightarrow 12r = 5n + 18 \quad \dots\dots(2)$$

Solving (1) and (2), we get $n = 6$

Q2. Coefficient of x^{11} in the expansion of $(1+x^2)^4 (1+x^3)^7 (1+x^4)^{12}$ is

(A) 1051 (B) 1106 (C) 1113 (D) 1120

Ans. C

Sol. Coefficient of $x^{11} \equiv \frac{(1+x^2)^4 (1+x^3)^7 (1+x^4)^{12} (1-x^2)^4}{(1-x^2)^4}$

$$\begin{aligned} \text{Coefficient of } x^{11} &\equiv (1-x^8)^4 (1+x^4)^8 (1+x^3)^7 (1-x^2)^{-4} \\ &= (1-4x^8) (1+x^4)^8 (7x^3+35x^9) (1-x^2)^{-4} \\ &= (7x^3+35x^9-28x^{11}) (1+x^4). (1-x^2)^{-4} \end{aligned}$$

$$\begin{aligned} \text{Coefficient of } x^8 &= (7x+35x^6-28x^8) (1+8x^4+28x^8) (1-x^2)^{-4} \\ &= (7+35x^6-28x^8+56x^4+196x^8) (1-x^2)^{-4} \end{aligned}$$

$$\begin{aligned} \text{Coefficient of } t^4 &\equiv (7+56t^2+35t^3+168t^4) (1-t)^{-4} \\ &= 7 \cdot {}^7C_3 + 56 \cdot {}^5C_3 + 35 \cdot {}^4C_3 + 168 \\ &= 245 + 700 + 168 = 1113. \end{aligned}$$

Q3. The coefficient of x^9 in the expansion of $(1+x)(1+x^2)(1+x^3)\dots(1+x^{100})$ is :

Ans. 8

Sol. $9 = (0, 9) (1, 8), (2, 7), (3, 6), (4, 5)$ # 5 cases

$9 = (1, 2, 6), (1, 3, 5), (2, 3, 4)$ # 3 cases

total = 8

Q4. Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1+x)^2 + (1+x)^3 + \dots + (1+x)^{49} + (1+mx)^{50}$ is $(3n+1) {}^{51}C_3$ for some positive

integer n. Then the value of n is

Ans. 5

Sol. ${}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^{49}C_2 + {}^{50}C_2 m^2 = (3n + 1) {}^{51}C_3$
 $\Rightarrow {}^{50}C_3 + {}^{50}C_2 m^2 = (3n + 1) {}^{51}C_3$
 $\Rightarrow {}^{50}C_3 + {}^{50}C_2 + (m^2 - 1) {}^{50}C_2 = 3n {}^{51}C_3 + {}^{51}C_3$
 $\Rightarrow n = \frac{m^2 - 1}{51}$
 $\Rightarrow m^2 = 51n + 1$ must be a perfect squared
 $\Rightarrow n = 5$ and $m = 16$

Q5. Let $X = ({}^{10}C_1)^2 + 2({}^{10}C_2)^2 + 3({}^{10}C_3)^2 + \dots + 10({}^{10}C_{10})^2$,
 where ${}^{10}C_r, r \in \{1, 2, \dots, 10\}$ denote binomial coefficients. Then, the value of $\frac{1}{1430} X$
 is _____

Ans. 646

Sol. $x = \sum_{i=1}^{10} r \cdot {}^{10}C_r = 10 \cdot \sum_{i=1}^{10} {}^9C_{r-1} \cdot {}^{10}C_{10-r} = 10 \cdot {}^{19}C_9$
 Now $\frac{x}{1430} = \frac{10 \cdot {}^{19}C_9}{1430} = \frac{{}^{19}C_9}{11 \times 13} = \frac{19 \cdot 17 \cdot 16}{8} = 19 \times 34 = 646$

Q6. For nonnegative integers s and r, let

$$\binom{s}{r} = \begin{cases} \frac{s!}{r!(s-r)!} & \text{if } r \leq s \\ 0 & \text{if } r > s \end{cases}$$

For positive integers m and n, let

$$g(m, n) = \sum_{p=0}^{m+n} \frac{f(m, n, p)}{\binom{n+p}{p}}$$

where for any nonnegative integers p,

$$f(m, n, p) = \sum_{i=0}^p \binom{m}{i} \binom{n+i}{p} \binom{p+n}{p-i}$$

Then which of the following statements is / are **TRUE** ?

- (A) $g(m, n) = g(n, m)$ for all positive integers m, n
- (B) $g(m, n+1) = g(m+1, n)$ for all positive integers m, n
- (C) $g(2m, 2n) = 2g(m, n)$ for all positive integers m, n
- (D) $g(2m, 2n) = (g(m, n))^2$ for all positive integers m, n

Ans. A, B, D

Sol.

$$\begin{aligned}
 f(m, n, p) &= \sum_{i=0}^p \left({}^m C_i \frac{(n+i)!}{(n+i-p)!p!} \times \frac{(p+n)!}{(n+i)!(p-i)!} \right) \\
 &= \sum_{i=0}^p {}^m C_i \left(\frac{n!}{(n+i-p)!(p-i)!} \right) \left(\frac{p+n!}{(n)!(p)!} \right) \\
 &= {}^{n+p} C_p \sum_{i=0}^p {}^m C_i {}^n C_{p-i} \\
 f(m, n, p) &= {}^{n+p} C_p \left({}^m C_0 {}^n C_p + {}^m C_1 {}^n C_{p-1} + \dots + {}^m C_p {}^n C_0 \right) \\
 &= {}^{n+p} C_p \cdot {}^{m+n} C_p \\
 g(m, n) &= \sum_{p=0}^{n+m} \frac{{}^{n+p} C_p \cdot {}^{m+n} C_p}{{}^{n+p} C_p} = 2^{m+n}
 \end{aligned}$$

Q7. Let a and b be two nonzero real numbers. If the coefficient of x^5 in the expansion of $\left(ax^2 + \frac{70}{27bx}\right)^4$ is equal to the coefficient of x^{-5} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^7$, then the value of 2b is _____.

Ans. 3

Sol.

$$\begin{aligned}
 T_{r+1} &= {}^4 C_r (a \cdot x^2)^{4-r} \cdot \left(\frac{70}{27bx} \right)^r \\
 &= {}^4 C_r \cdot a^{4-r} \cdot \frac{70^r}{(27b)^r} \cdot x^{8-3r}
 \end{aligned}$$

$$\text{here } 8 - 3r = 5$$

$$8 - 5 = 3r \Rightarrow r = 1$$

$$\therefore \text{coeff.} = 4 \cdot a^3 \cdot \frac{70}{27b}$$

$$\begin{aligned}
 T_{r+1} &= {}^7 C_r (ax)^{7-r} \left(\frac{-1}{bx^2} \right)^r \\
 &= {}^7 C_r \cdot a^{7-r} \left(\frac{-1}{b} \right)^r \cdot x^{7-3r}
 \end{aligned}$$

$$7 - 3r = -5 \Rightarrow 12 = 3r \Rightarrow r = 4$$

$$\text{Coeff. : } {}^7 C_4 \cdot a^3 \cdot \left(\frac{-1}{b} \right)^4 = \frac{35a^3}{b^4}$$

$$\text{Now } \frac{35a^3}{b^4} = \frac{280a^3}{27b}$$

$$b^3 = \frac{35 \times 27}{280} = b = \frac{3}{2} \Rightarrow 2b = 3$$

Exercise C1

Q1. The coefficient of x^{53} in the expansion $\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$ is

- (A) ${}^{100}C_{47}$ (B) ${}^{100}C_{53}$ (C) $-{}^{100}C_{53}$ (D) $-{}^{100}C_{100}$

Ans. C

Sol. $S = \sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$

$$S = {}^{100}C_0 (x-3)^{100} + {}^{100}C_1 (x-3)^{99} \cdot 2 + \dots + {}^{100}C_{100} \cdot 2^{100}$$

$$S = (2 + (x-3))^{100} = (x-1)^{100}$$

$$\text{Co-efficient of } x^{52} = {}^{100}C_{52} = {}^{100}C_{48}$$

Q2. If the sum of the coefficients in the expansion of $\left(x^2 + \frac{2}{x^3}\right)^n$ is 243, then the term independent of x , is equal to

- (A) 11 (B) $\frac{7}{2}$ (C) 15 (D) 40

Ans. D

Sol. -

Q3. The value of $4 \{ {}^nC_1 + 4 \cdot {}^nC_2 + 4^2 \cdot {}^nC_3 + \dots + 4^{n-1} \}$ is

- (A) 0 (B) $5^n + 1$ (C) 5^n (D) $5^n - 1$

Ans. D

Sol. -

Q4. The value of $\sum_{r=0}^n \frac{1}{(2r)!(2n-2r)!}$, is equal to

- (A) $\frac{2^{2n}}{(2n)!}$ (B) $\frac{2^{2n-1}}{(2n)!}$ (C) $\frac{2^{n-1}}{n!}$ (D) $\frac{2^n}{n!}$

Ans. B

Sol. $(\tan 18^\circ)(\tan 70^\circ)(\tan 108^\circ)(\tan 162^\circ) = 1$

hence we are left with

$$(-)(\sin 36^\circ)(\cos 54^\circ)(\cos 126^\circ)(\sin 144^\circ)$$

$$(-)(\sin 36^\circ)(\sin 36^\circ)(-\sin 36^\circ) = \sin^4 36^\circ$$

$$= \left(\frac{1 - \cos 72^\circ}{2} \right)^2 = \frac{(1 - \sin 18^\circ)^2}{4}$$

$$= \frac{1}{4} \left(1 - \frac{\sqrt{5}-1}{4} \right) = \frac{1}{4} \left(\frac{5-\sqrt{5}}{4} \right)^2 = \frac{5}{4} \left(\frac{\sqrt{5}-1}{4} \right)^2 = \frac{5}{4} \sin^2 18^\circ \Rightarrow k = \frac{5}{4} \text{ Ans.}$$

Q5. In the expansion of $(1+x)^n (1+y)^n (1+z)^n$, the sum of the co-efficients of the terms of degree 'r' is

- (A) ${}^n C_r$ (B) ${}^n C_{r^3}$ (C) ${}^{3n} C_r$ (D) $3 \cdot {}^{2n} C_r$

Ans. C

Sol. Put $x = y = z$, we get $(1+x)^{3n}$

$\therefore$ coeff. of terms of degree $r = {}^{3n} C_r$

Q6. The sum of the last n coefficients in the expansion of $(1+x)^{2n-1}$ when expanding in ascending power of x, is equal to

- (A) 2^{2n-1} (B) 2^{2n-2} (C) 2^{2n} (D) $2^{2n} - 2^n$

Ans. B

Sol. -

Q7. The fractional part of $\frac{3^{2n}}{8}$, is equal to

- (A) $\frac{3}{8}$ (B) $\frac{7}{8}$ (C) $\frac{1}{8}$ (D) None of these

Ans. C

Sol. -

Q8. Given positive integers $r > 1$, $n > 2$, and the coefficients of the $(3r)^{\text{th}}$ and $(r+2)^{\text{th}}$ terms in the binomial expansion of $(1+x)^{2n}$ are equal, which, if any, of the following statements is true ?

- (A) $n = 2r$ (B) $n = 3r$ (C) $n = 2r + 1$ (D) None of these

Ans. A

Sol. -

Q9. If in the expansion of $(1+x)^m (1-x)^n$, coefficients of x and x^2 are 3 and -6 respectively, then m is

- (A) 6 (B) 9 (C) 12 (D) 24

Ans. C

Sol. Consider $(1+x)^m (1-x)^n$

$$\Rightarrow [{}^m C_0 + {}^m C_1 x + {}^m C_2 x^2 + \dots \dots \dots {}^m C_m x^m] [{}^n C_0 + {}^n C_1 (-x) + {}^n C_2 (-x)^2 + \dots \dots \dots {}^n C_n (-x)^n]$$

$$[\because (1+x)^n = {}^nC_0 + {}^nC_1x + \dots + {}^nC_nx^n]$$

On multiplying and comparing

$$\text{Now coefficient of } x = {}^mC_0 \cdot {}^nC_1(-1) + {}^mC_1 \cdot {}^nC_0 = m - n = 3$$

$$\text{Coeff. of } x^2 = {}^mC_0 \cdot {}^nC_2(-1)^2 + {}^mC_2 \cdot {}^nC_0 + {}^nC_1(-1)$$

$$\begin{aligned} x^2 &= {}^mC_0 \cdot {}^nC_2(-1)^2 + {}^mC_2 \cdot {}^nC_0 + {}^nC_1(-1) \\ &= \frac{n(n-1)}{2} + \frac{m(m-1)}{2} - mn = -6 \\ &= \frac{n(n-1)}{2} + \frac{(3+n)(3+n-1)}{2} - (3+n) = -6 \\ &= \frac{n^2 - n}{2} + \frac{n^2 + 5n + 6}{2} - 3n - n^2 = -6 \\ &= \frac{n^2 - n + n^2 + 5n + 6 - 6n - 2n^2}{2} = -6 \\ &= -2n + 6 = -12 \end{aligned}$$

$$n! = 9 \quad \text{Now } m = 3 + n = 12 \quad \text{Ans.}$$

Q10. ${}^{n-1}C_r = (K^2 - 3) \cdot {}^nC_{r+1}$, if $K \in$

(A) $[-\sqrt{3}, \sqrt{3}]$ (B) $(-\infty, -2)$ (C) $(2, \infty)$ (D) $(\sqrt{3}, 2]$

Ans. D

Sol. ${}^{n-1}C_r = (k^2 - 3) \cdot {}^nC_{r+1}$

$$\frac{(n-1)!}{r!(n-r-1)!} = (K^2 - 3) \cdot \frac{n!}{(r+1)!(n-r-1)!}$$

$$\Rightarrow \frac{r+1}{n} + 3 = k^2 \Rightarrow k^2 - 3 > 0$$

$$\Rightarrow k \in (-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty) \dots (i)$$

$$n-1 \geq r$$

$$\frac{r+1}{n} \leq 1 \Rightarrow k^2 \leq 3+1$$

$$-2 \leq k \leq 2 \dots (ii)$$

$$k \in [-2, \sqrt{3}) \cup (\sqrt{3}, 2]$$

Q11. The value of $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} \dots + \binom{30}{20}\binom{30}{30}$ is, where $\binom{n}{r} = {}^nC_r$.

(A) $\binom{30}{10}$ (B) $\binom{30}{15}$ (C) $\binom{60}{30}$ (D) $\binom{31}{10}$

Ans. A

Sol. -

Q12. The polynomial $1 - x + x^2 - x^3 + \dots + x^{16} - x^{17}$ may be written in the form $\sum_{i=0}^{17} a_i y^i$ where $y = x + 1$ and a_i 's are constants. Then a_2 equals –

- (A) The coefficient of t^3 in the expansion of $1 - (1 - t)^{18}$ (B) $\sum_{n=2}^{17} \binom{n}{2}$ (C) 816
(D) The coefficient of t^3 in the expansion of $1 - (1 - t)^{17}$

Ans. C, A, B

Sol. -

Q13. In the expansion of $\left(x^3 + 3.2^{-\log\sqrt{2}\sqrt{x^3}}\right)^{11}$

- (A) there appears a term with the power x^2
(B) there does not appear a term with the power x^2
(C) there appears a term with the power x^{-3}
(D) the ratio of the co-efficient of x^3 to that of x^{-3} is $1/3$

Ans. D, C, B

Sol. -

Q14. If it is known that the third term of the binomial expansion $(x + x^{\log_{10} x})^5$ is 10^6 then x is equal to

- (A) 10 (B) $10^{-5/2}$ (C) 100 (D) 5

Ans. A, B

Sol. -

Q15. The expansion of $\left(\sqrt{x} + \frac{1}{2\sqrt[4]{x}}\right)^n$ is arranged in decreasing power of x . If coefficient of first three terms form an A.P. then in expansion, the integral powers of x are :

- (A) 0 (B) 2 (C) 4 (D) 8

Ans. B

Sol. -

Q16. $\sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n \binom{n}{i} \binom{n}{j} \binom{n}{k}$, where $\binom{n}{r} = {}^nC_r$:

- (A) is less than 5000 if $n = 3$ (B) is greater than 600 if $n = 3$

(C) is less than 5000 if $n = 4$ (D) is greater than 4000 if $n = 4$

Ans. D, C, A

Sol. -